50Days-coding-challenge

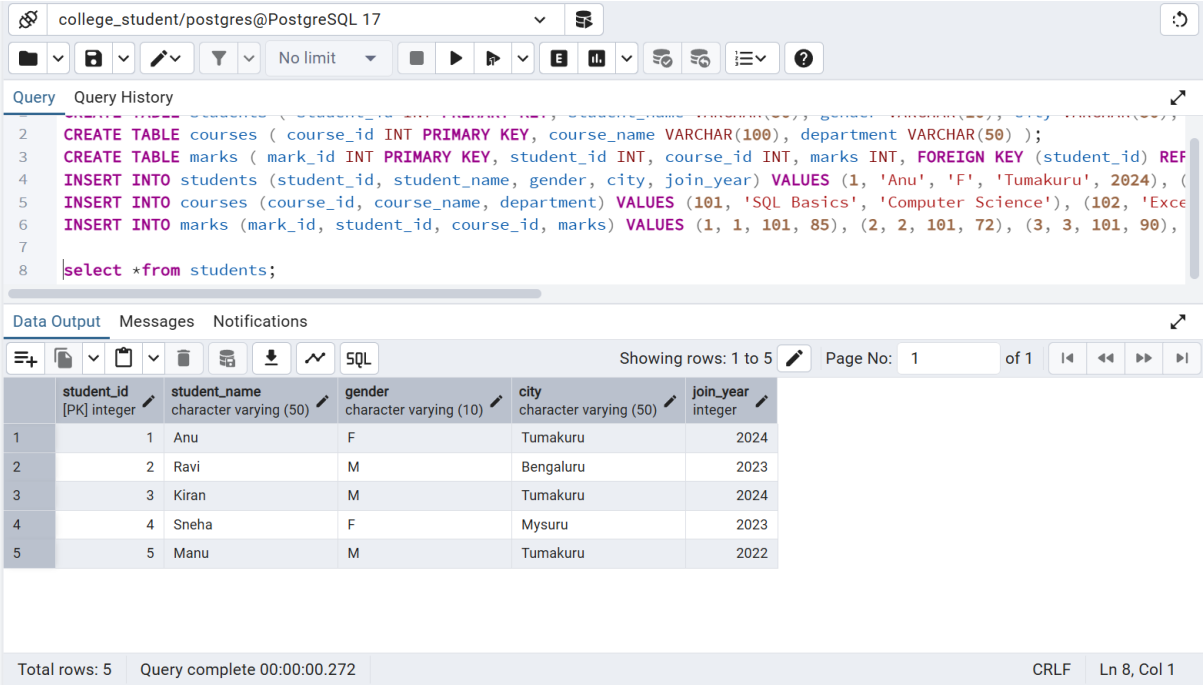
Day-15[SQL]

College Student Management System

Name:M NITHIN

Level 1: Very Basic

1. Display all students.



	student_id [PK] integer	student_name character varying (50)	gender character varying (10)	city character varying (50)	join_year integer
1	1	Anu	F	Tumakuru	2024
2	2	Ravi	M	Bengaluru	2023
3	3	Kiran	M	Tumakuru	2024
4	4	Sneha	F	Mysuru	2023
5	5	Manu	M	Tumakuru	2022

2. Display only student_name and city from students table.

college_student/postgres@PostgreSQL 17

Query | Query History

```
INSERT INTO students (student_id, student_name, gender, city, join_year) VALUES (1, 'Anu', 'F', 'Tumakuru', 2024),
INSERT INTO courses (course_id, course_name, department) VALUES (101, 'SQL Basics', 'Computer Science'), (102, 'Exc
INSERT INTO marks (mark_id, student_id, course_id, marks) VALUES (1, 1, 101, 85), (2, 2, 101, 72), (3, 3, 101, 90),

select *from students;

select student_name,city from students;
```

Data Output | Messages | Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	student_name character varying (50)	city character varying (50)
1	Anu	Tumakuru
2	Ravi	Bengaluru
3	Kiran	Tumakuru
4	Sneha	Mysuru
5	Manu	Tumakuru

Total rows: 5 | Query complete 00:00:00.239 | CRLF | Ln 10, Col 40

3. Show all courses.

college_student/postgres@PostgreSQL 17

Query | Query History

```
INSERT INTO marks (mark_id, student_id, course_id, marks) VALUES (1, 1, 101, 85), (2, 2, 101, 72), (3, 3, 101, 90),

select *from students;

select student_name,city from students;

select*from courses;
```

Data Output | Messages | Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	course_id [PK] integer	course_name character varying (100)	department character varying (50)
1	101	SQL Basics	Computer Science
2	102	Excel for Analysts	Commerce
3	103	Statistics	Mathematics

Total rows: 3 | Query complete 00:00:00.222 | CRLF | Ln 12, Col 1

4. Display students who are from Tumakuru.

college_student/postgres@PostgreSQL 17

Query | Query History

```
select student_name,city from students;

select*from courses;

select *from students
where city='Tumakuru';
```

Data Output | Messages | Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	student_id [PK] integer	student_name character varying (50)	gender character varying (10)	city character varying (50)	join_year integer
1	1	Anu	F	Tumakuru	2024
2	3	Kiran	M	Tumakuru	2024
3	5	Manu	M	Tumakuru	2022

Total rows: 3 | Query complete 00:00:00.239 | CRLF | Ln 15, Col 23

5. Display students who joined in 2024.

college_student/postgres@PostgreSQL 17

Query | Query History

```
select*from courses;

select *from students
where city='Tumakuru';

select*from students
where join_year=2024;
```

Data Output | Messages | Notifications

Showing rows: 1 to 2 Page No: 1 of 1

	student_id [PK] integer	student_name character varying (50)	gender character varying (10)	city character varying (50)	join_year integer
1	1	Anu	F	Tumakuru	2024
2	3	Kiran	M	Tumakuru	2024

Total rows: 2 | Query complete 00:00:00.216 | CRLF | Ln 17, Col 1

Level 2: Filtering

1. Show students whose gender is F.

Query Query History

```
where city='Tumakuru';

select*from students
where join_year=2024;

select*from students
where gender='F';
```

Data Output Messages Notifications

Showing rows: 1 to 2 Page No: 1 of 1

	student_id [PK] integer	student_name character varying (50)	gender character varying (10)	city character varying (50)	join_year integer
1	1	Anu	F	Tumakuru	2024
2	4	Sneha	F	Mysuru	2023

Successfully run. Total query runtime: 232 msec. 2 rows affected.

Total rows: 2 Query complete 00:00:00.232 CRLF Ln 20, Col 1

2. Show marks greater than 80.

college_student/postgres@PostgreSQL 17

No limit

Query Query History

```
where join_year=2024;

select*from students
where gender='F';

select * from marks
where marks>80;
```

Data Output Messages Notifications

Showing rows: 1 to 4 Page No: 1 of 1

	mark_id [PK] integer	student_id integer	course_id integer	marks integer
1	1	1	101	85
2	3	3	101	90
3	4	4	102	88
4	7	2	102	81

Total rows: 4 Query complete 00:00:00.235 CRLF Ln 23, Col 1

3. Display course names from Commerce department.

college_student/postgres@PostgreSQL 17

Query | Query History

```
select * from marks
where marks>80;

select course_name
from courses
where department='Commerce'
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	course_name character varying (100)
1	Excel for Analysts

Total rows: 1 | Query complete 00:00:00.186 | CRLF | Ln 26, Col 1

4. Show students who are not from Bengaluru.

college_student/postgres@PostgreSQL 17

Query | Query History

```
select course_name
from courses
where department='Commerce'

select * from students
where city<>'Bengaluru';
```

Data Output | Messages | Notifications

Showing rows: 1 to 4 Page No: 1 of 1

	student_id [PK] integer	student_name character varying (50)	gender character varying (10)	city character varying (50)	join_year integer
1	1	Anu	F	Tumakuru	2024
2	3	Kiran	M	Tumakuru	2024
3	4	Sneha	F	Mysuru	2023
4	5	Manu	M	Tumakuru	2022

Total rows: 4 | Query complete 00:00:00.227 | CRLF | Ln 30, Col 10

5. Display marks between 70 and 90.

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Query | Query History

```
where department='Commerce'

select * from students
where city<>'Bengaluru';

SELECT * FROM marks
WHERE marks >=70 and marks<=90;
```

Data Output | Messages | Notifications

Showing rows: 1 to 6 Page No: 1 of 1

	mark_id [PK] integer	student_id integer	course_id integer	marks integer
1	1	1	101	85
2	2	2	101	72
3	3	3	101	90
4	4	4	102	88
5	6	1	103	79
6	7	2	102	81

Total rows: 6 Query complete 00:00:00.371 CRLF Ln 33, Col 1

Level 3: Sorting

1. Display all students ordered by student_name ascending.

college_student/postgres@PostgreSQL 17

Query | Query History

```
where city<>'Bengaluru';

SELECT * FROM marks
WHERE marks >=70 and marks<=90;

select*from students
order by student_name asc;
```

Data Output | Messages | Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	student_id [PK] integer	student_name character varying (50)	gender character varying (10)	city character varying (50)	join_year integer
1	1	Anu	F	Tumakuru	2024
2	3	Kiran	M	Tumakuru	2024
3	5	Manu	M	Tumakuru	2022
4	2	Ravi	M	Bengaluru	2023
5	4	Sneha	F	Mysuru	2023

Total rows: 5 Query complete 00:00:00.230 CRLF Ln 36, Col 1

2. Show marks ordered from highest to lowest.

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Query | Query History

```
select*from students
order by student_name asc;

select*from marks
order by marks desc;
```

Data Output | Messages | Notifications

Showing rows: 1 to 7 Page No: 1 of 1

	mark_id [PK] integer	student_id integer	course_id integer	marks integer
1	3	3	101	90
2	4	4	102	88
3	1	1	101	85
4	7	2	102	81
5	6	1	103	79
6	2	2	101	72
7	5	5	103	67

Total rows: 7 Query complete 00:00:00.252 CRLF Ln 41, Col 1

3. Display students ordered by join_year descending.

Query | Query History

```
order by student_name asc;

select*from marks
order by marks desc;

select*from students
order by join_year desc;
```

Data Output | Messages | Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	student_id [PK] integer	student_name character varying (50)	gender character varying (10)	city character varying (50)	join_year integer
1	1	Anu	F	Tumakuru	2024
2	3	Kiran	M	Tumakuru	2024
3	2	Ravi	M	Bengaluru	2023
4	4	Sneha	F	Mysuru	2023
5	5	Manu	M	Tumakuru	2022

Total rows: 5 Query complete 00:00:00.277 CRLF Ln 41, Col 1

Level 4: Aggregate Basics

1. Find total number of students.

college_student/postgres@PostgreSQL 17

Query | Query History

```
order by marks desc;

select*from students
order by join_year desc;

select count(*) as total_number_of_students from students;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	total_number_of_students bigint
1	5

Total rows: 1 | Query complete 00:00:00.189 | CRLF | Ln 45, Col 59

2. Find average marks.

college_student/postgres@PostgreSQL 17

Query | Query History

```
select*from students
order by join_year desc;

select count(*) as total_number_of_students from students;

select avg(marks) as Avg_marks from marks;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	avg_marks numeric
1	80.2857142857142857

Total rows: 1 | Query complete 00:00:00.148 | CRLF | Ln 48, Col 1

3. Find highest marks.

college_student/postgres@PostgreSQL 17

No limit

Query | Query History

```
45  select count(*) as total_number_of_students from students;
46
47  select avg(marks) as Avg_marks from marks;
48
49  select max(marks) as Highest_marks from marks;
50
51
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	highest_marks integer
1	90

Total rows: 1 Query complete 00:00:00.142 CRLF Ln 49, Col 1

4. Find lowest marks.

college_student/postgres@PostgreSQL 17

No limit

Query | Query History

```
45  select count(*) as total_number_of_students from students;
46
47  select avg(marks) as Avg_marks from marks;
48
49  select max(marks) as Highest_marks from marks;
50
51  select min(marks) as lowest_marks from marks;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	lowest_marks integer
1	67

Total rows: 1 Query complete 00:00:00.178 CRLF Ln 51, Col 1

5. Find total marks scored by all students.

college_student/postgres@PostgreSQL 17

No limit

Query | Query History

```sql
select avg(marks) as Avg_marks from marks;

select max(marks) as Highest_marks from marks;

select min(marks) as lowest_marks from marks

select sum(marks) as Total_marks from marks;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	total_marks bigint
1	562

Total rows: 1 | Query complete 00:00:00.205 | CRLF | Ln 54, Col 1